Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[18CE54]**

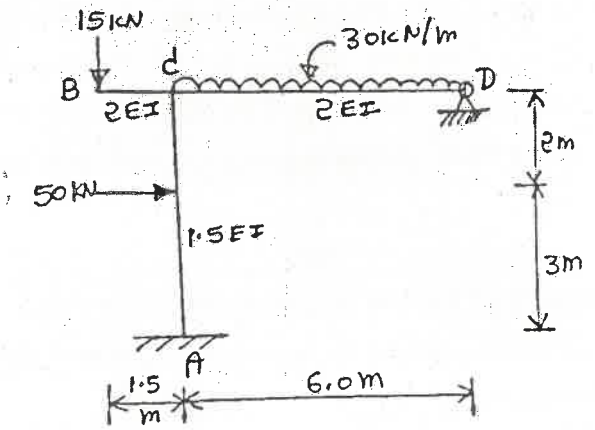
FIFTH SEMESTER B.Tech EXAMINATION, MARCH–2022

STRUCTURAL ANALYSIS-II

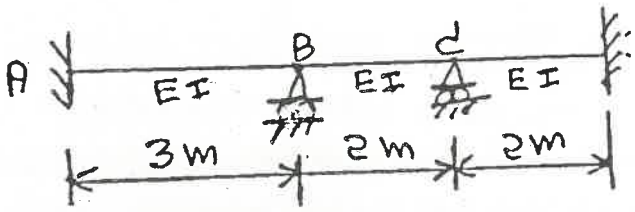
Time:3 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

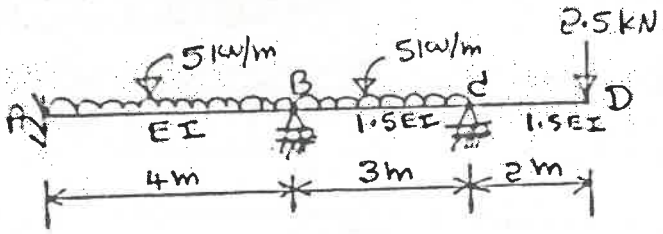
1. Analyze the rigid jointed plane frame loaded as shown below by slope deflection method and hence determine its bending moment diagram. **14**



2. Analyze the continuous beam as shown below by moment distribution method due to the sinking effect of support B by 30 mm and hence determine its bending moment diagram. Consider the values of young's modulus = $2X10^5$ N/mm^2 for beam material and moment of inertia of the beam cross section = $3X10^6$ mm^4. **14**



3. Analyze the continuous beam loaded as shown below by Kani's method and hence determine its bending moment diagram. **14**



OR

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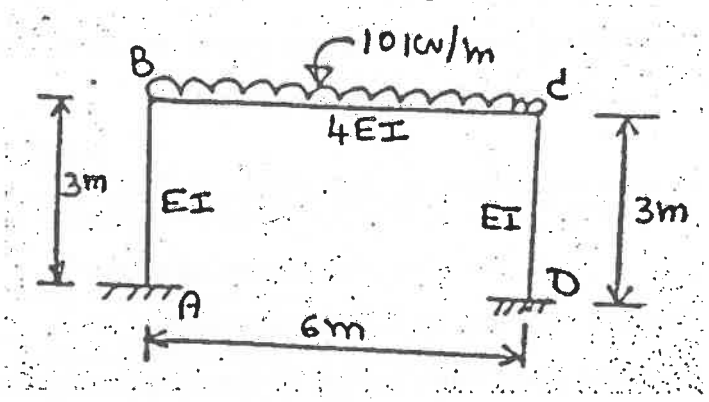
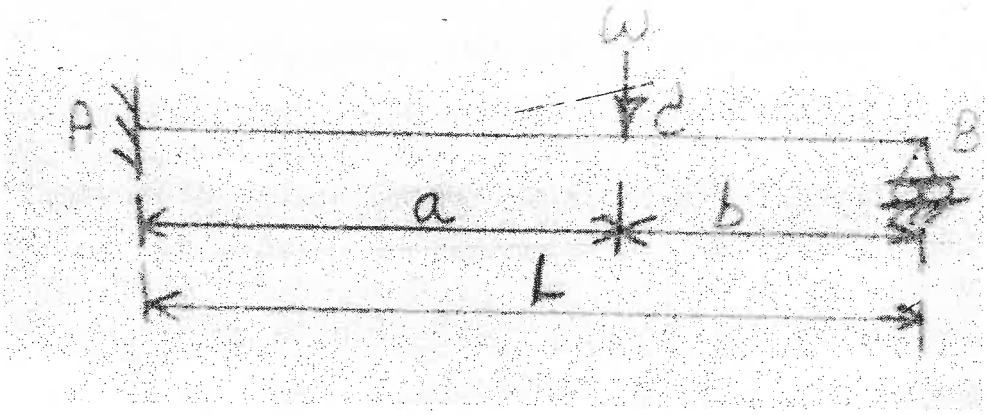
4. Analyze the rigid jointed plane frame loaded as shown below by Kani's method and hence determine its bending moment diagram. **14**



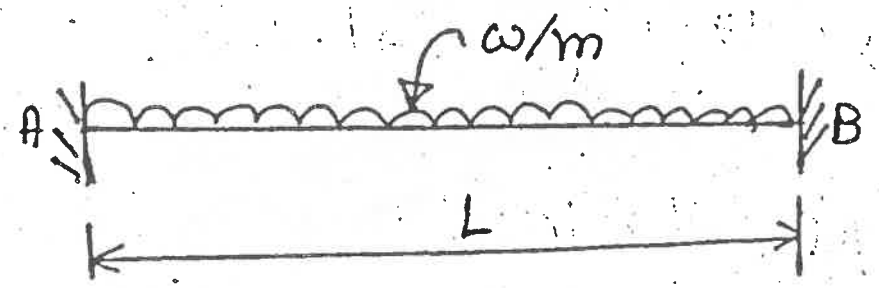
5. Determine the elastic section modulus, plastic section modulus, shape factor and plastic moment capacity with respect to major axis for Indian rolled steel I section ISMB 200 having following details. Depth of the section = 200mm, flange width = 100 mm, flange thickness = 10.80 mm, web thickness = 5.70 m and web depth = 178.40 mm. Consider the yield strength of the steel as 250 N/mm^2. **14**

OR

6. a. Determine the collapse load for the propped cantilever beam loaded as shown below by static equilibrium method of approach. Consider the plastic moment capacity of the beam material as M_P. **07**



b. Determine the collapse load for the fixed beam loaded as shown below by static equilibrium method of approach. Consider the plastic moment capacity of the beam material as M_P. **07**

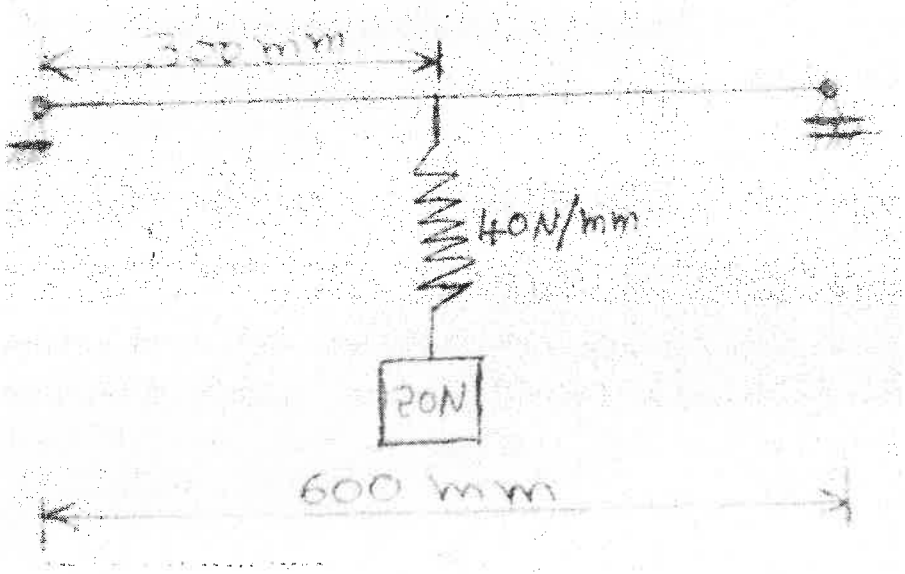


7. a. Define the following. **04**

i. Degree of freedom

ii. Natural frequency.

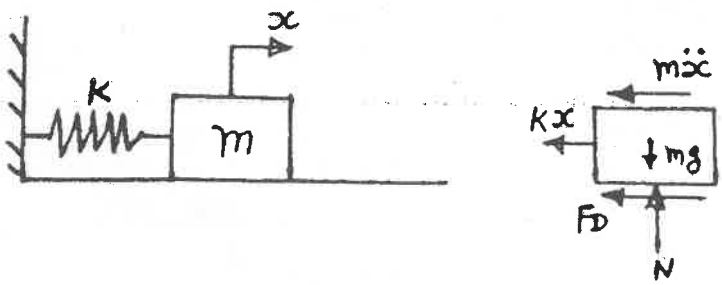
b. Determine the natural frequency of the system as shown in the figure below. The mass of the beam may be neglected. Consider the young's modulus of the material as 2.10×10^5 N/mm^2. The cross section dimension of the beam with rectangular cross section is width = 100 mm and depth = 150 mm. **10**



OR

8. a. Explain briefly about. **4**

i. Free and forced vibration

ii. Different damping systems.

b. For the single degree freedom with coulomb damping along with free body diagram as shown in the figure below, determine the coefficient of friction if a tensile force P of magnitude (mg) elongates the spring by 6.00 mm. The initial amplitude of x_0 = 600 mm reduces by 0.80 times of its value after 20 cycles by considering the amplitude reduction in each cycle as $(4 \times F_D)/K$. **10**